LLM-Empowered Autonomous Edge AI for Anomaly Detection and Resolution in Distributed Systems

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***Abstract*—There is a growing need for strong security measures to combat cyberattacks with increasing data consumption and reliance on interconnected networks. This is especially required at the network edge, where the data are present and first acquired. Cyber attacks such as distributed denial-of-service (DDoS) attacks and others pose risk and various other inconveniences to all stakeholder involved. The dynamic nature of these attacks and the lack of processing power at the edge require new methods to tackle cyber-attacks. This paper proposes a novel approach for network anomaly detection and mitigation that uses edge AI and the contextual understanding power of Large Language Models (LLMs). The system utilizes deep learning methods to identify cyberattacks where the models are deployed on edge devices and servers enabling quick identification of anomalies. The paper then explores the use of LLMs to automate the resolution and mitigation of these attacks in real-time, leading to lower impact by reducing downtimes and increasing availability. This research aims to contribute to the field of network security by proposing an edge AI driven solution for real-time local DDoS attack detection and the utilization of LLMs for network anomaly resolution and system adaptation.**

***Index Terms*—Distributed Denial of Service, Edge AI, Large Language Models, GPT, Deep Learning, Anomaly Detection, Network Security**

I. Introduction

Strong security measures are required due to the constant nature of cyberspace, particularly at the network edge, where data is first generated and collected. Attacks like distributed denial of service (DDoS) pose a serious risk at the edge because they flood edge devices with malicious traffic, disrupting essential services closer to the point of origin. These attacks have the potential to seriously damage industrial control systems, edge-based applications, and Internet of Things (IoT) networks, resulting in large-scale financial losses and reduced operational effectiveness. Traditional techniques for detecting and mitigating edge-based threats frequently depend on centralized processing, which can cause delays and restrict response times. This research proposes a novel solution to the problem of DDoS attacks at the network edge utilizing Edge AI.

Edge AI refers to the deployment and execution of AI/ML algorithms and models on edge devices, which are the physical endpoints of a typical network where data are generated and initially processed. This approach allows us to perform real-time processing, leading to reduced latency, enhanced privacy, and improved efficiency by performing computations closer to the data source. In terms of network security, Edge AI allows for faster anomaly detection and resolution with minimal delays compared to a server-only-based approach.

For the goal of detecting network anomalies in edge devices in real time, this study suggests an innovative edge AI system. Using lightweight deep learning (DL) models in edge devices, which were trained on a well established benchmark dataset, CICDDoS2019, the system is able to detect malicious activity linked to DDoS attacks by analyzing patterns in network data. By deploying these models directly on edge devices, they provide local anomaly detection and real-time analysis. Some of the benefits of this strategy include reducing latency for faster mitigation, increasing efficiency by freeing up network traffic, improving scalability to support more edge devices, and protecting privacy by processing sensitive data locally on edge devices.

This research also explores the potential of Large Language Models (LLMs) such as GPT for further enhancements to the system. GPT's natural language understanding, planning, and coding abilities could potentially be used to automate resolution techniques and give the system real-time adaptability to novel attack patterns. This research adds to the current effort to improve network security by proposing a novel Edge AI-based strategy for real-time detection of DDoS attacks and establishing the framework for further investigation of LLM integration in network anomaly resolution.

II. Related Work

Shen et al. [1] present a framework for an autonomous Edge AI system that makes use of LLMs, specifically GPT, to manage the use of current AI models, train new ones at the network edge, and solve problems with understanding

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user requests, executing AI models, and producing code. Their visual question answering (VQA), image captioning, and image classification tests demonstrate that GPT can produce code for the training of the federated learning model. This makes it possible to continuously increase performance and provides the foundation for the development of an autonomous system that can improve itself over time while ensuring user privacy. Feature selection methods have been researched in a number of studies to improve the performance of AI-based DDoS attack detection systems. We use the CICDDoS2019 dataset, which is also used by Raza et al. [2] and Shaikh et al. [3]. These studies identify and remove features that are not relevant, such as properties related to networks, columns that contain a high percentage of zero values, and features with low variability. This approach allows optimizing the training process and may improve model performance. Chartuni and Marquez [4] present a DDoS attack detection system employing a multiclassifier neural network architecture built on the CICDDoS2019 dataset. Their model uses a seven-layer sequential structure: an input layer, five hidden levels with ReLU activation functions, and an output layer at the end. In addition, a dropout layer was added to avoid overfitting. Based on the way in which the equivalent distinct types of attack are in the feature space, the authors investigated three training situations. Their model achieves an accuracy of 94.57%.

Qiang Fu et al. [6] presents a method for the detection of execution anomalies in distributed systems by converting raw unstructured logs into log keys and using clustering techniques to identify frequency patterns. Their automated approach showed an improvement in fault detection in distributed environments. Thwaini MH et al. [7] the study focuses on techniques that were non-invasive, it evaluates different models, such as decision trees and random forests. It also highlights unresolved issues that need to be addressed to improve such systems. Ali Shamekhi et al. [8] propose a DDoS attack detection method that clusters the behavior of the network user based on adaptive time intervals. The paper contributes by introducing the time stamp feature as an indicator of behavior. They were able to achieve higher accuracy and precision with the inclusion of this feature. Vedanth Ramanathan et al. [9] propose a novel CNN-based architecture for the detection of DDoS attacks on edge systems, the approach involves supervised learning. The paper was able to achieve an accuracy of 98.83% on unseen network traffic, while ensuring scalability.

Azar Abid Salih et al. [10] present a novel lightweight DL model, called Cybernet models. These are meant for the detection of DDoS attacks. They have a lower number of learnable parameters. They were able to achieve an accuracy of around 99.7% with the CICDDoS2019 dataset. Jahangir Shaikh et al. [11] propose a deep learning-based intrusion detection system that is a CNN-LSTM hybrid to extract spatial and temporal features. An autoencoder was used for dimensionality reduction. They were able to achieve an accuracy score of 99.86%. Muhammad Raza et al. [12] introduce a feature selection-based method to improve DDoS detection. They use filter-based, wrapper-based, and embedded feature selection approaches. This method was tested on five different models including NGBoost and the paper reported an improvement in detection accuracy.

Muder Almiani et al. [13] propose a novel method for the detection of DDoS attacks in Internet of Things (IoT) networks. It uses a deep Kalman backpropagation-based neural network, which enhances accuracy by filtering out noise from network traffic. The model achieves an accuracy of 94%. Peng Line et al. [14] focuses on developing a dynamic anomaly detection model, they use a Long Short Term Memory (LSTM) to build a Deep Neural Network (DNN) model and also use an Attention Mechanism (AM) to improve the model performance. They were able to achieve an accuracy of 96.2% with the CSE-CIC-IDS 2018 dataset. Mahmoud et al. [15] propose DDoSNet, a DDoS detection system in Software-Defined Networking (SDN) environments. They utilize Recurrent Neural Networks and combine them with an autoencoder. It can handle binary and multiclass classification in real time and can achieve an accuracy of around 99%. H Kim et al. [16] explore an AI-driven approach for detection of anomalies in Industrial smart manufacturing networks, for this they introduce a method that automates feature extraction and analyzes message intervals, patterns and network signatures and is able to identify any unusual network behaviors. They were able to achieve an F1 score of 0.99.

These studies provide useful information on the use of artificial intelligence and machine learning to detect DDoS attacks. Building on this foundation, our work proposes a novel Edge AI architecture that uses LLMs and client-edge co-inference to achieve real-time anomaly detection, faster mitigation techniques, and enhanced system efficiency.

III. Proposed Methodology

This section describes the proposed method for an autonomous Edge AI system utilizing Large Language Models (LLMs) and client-edge co-inference for real-time detection, mitigation, and resolution of DDoS attacks.

A. System Architecture

The proposed architecture utilizes Edge AI to perform real-time anomaly detection on edge devices in distributed networks. Edge devices and edge servers are connected using Message Queuing Telemetry Transport (MQTT). The ML Model works on a client-edge co-inference approach. This is done to reduce the load on the edge device, edge devices track network traffic and perform initial analysis, and send extracted features to the edge device for further analysis. The heavy-weight deep-learning model on the edge server performs the final anomaly detection. This co-inference approach ensures that there is less load on the edge device and also ensures low-latency and a higher accuracy for anomaly detection. Additionally, there is an LLM hosted on the cloud that is used for automation of the mitigation task, the LLM automated server configuration updates and allow for the mitigation of the anomaly. The automation tools are then used to reflect changes in the devices. Fig. 1 shows the architecture of the system.

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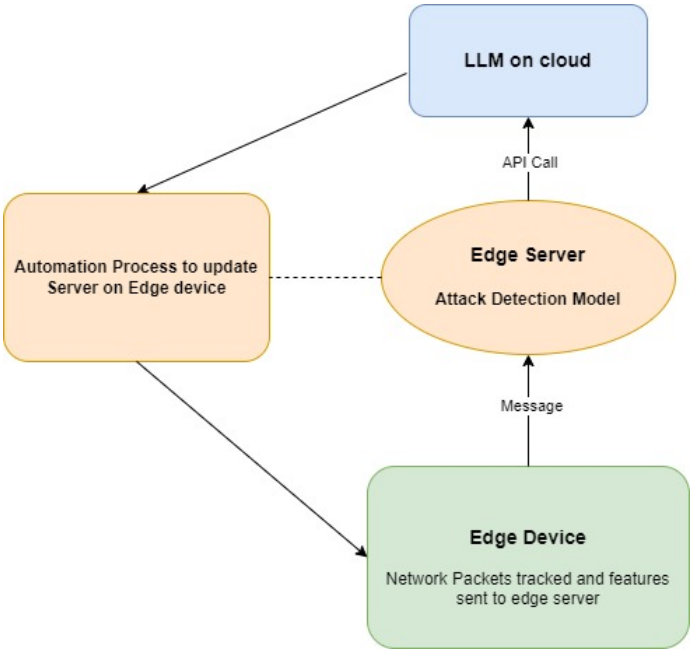

Fig. 1. System Architecture for Anomaly Detection.

B. Data Flow

The system has the following data flow:

- Network Traffic Capture and Initial Analysis: The edge devices constantly analyze all incoming network traffic and they process them and extract relevant features and perform lightweight analysis before transmitting them to the edge server.
- Feature Transmission: The extracted features are then sent via an MQTT broker to the edge server for final analysis to determine if there is any anomaly.
- Comprehensive Analysis: The heavyweight model on the edge server processes the received features and performs a final analysis, and if any anomaly is found, it then prompts the LLM with any necessary data.
- Mitigation and Automation: If an anomaly is detected, the LLM is then prompted. The LLM generates server configuration updates and sends them back to the edge server to be later sent to the edge device, which utilizes the new configurations and updates them on the local device.

C. Training Dataset

The dataset used by us is the CICDDoS2019 dataset [5], developed by the Canadian Institute for Cybersecurity (CIC). This is the primary training set for deep learning models; the dataset comprises a variety of DDoS attacks, such as reflection, amplification, and credential-stuffing attacks. Initially, the dataset contained 82 features. Through a comprehensive feature importance analysis and literature survey, we identified and selected 54 features that are most relevant while ensuring network independence. Fig. 2 illustrates the relative importance scores of the top 10 features in our analysis, where metrics like minimum packet length and forward packet length statistics were strong indicators of attack patterns. Network-specific features were intentionally omitted to ensure the model's generalizability across different types of network.

Three classes were considered, Benign, Syn Flood, and UDP Flood. The dataset was created by collecting the features of network packets from a controlled network on which attacks were performed using third-party software. The nature of the dataset, containing real network traffic patterns ensures that the models trained have an improved capacity to generalize and efficiently detect DDoS attacks in the real world. The features were extracted using the CICFlowMeter tool [17] [18], also developed by CIC. This is the same tool we used to extract features on edge devices, ensuring consistency in the features fed into the deep learning model.

```
Top 10 Important Features:
Min Packet Length: 0.1658
Fwd Packet Length Min: 0.1098
Total Length of Fwd Packets: 0.1093
Max Packet Length: 0.0966
Avg Fwd Segment Size: 0.0802
Fwd Packet Length Mean: 0.0721
Subflow Fwd Bytes: 0.0575
Average Packet Size: 0.0543
ACK Flag Count: 0.0436
Fwd Packet Length Max: 0.0398
```

Fig. 2. Feature Importance of the Top 10 Features.

D. Anomaly Detection Models & Deployment

Two machine learning models, a deep neural network and a random forest classifier, were tested. The Random Forest (RF) Classifier is an ensemble technique that constructs multiple decision trees during training and classifies based on the output of these trees. A hyperparameter search was performed to obtain the best possible hyperparameters. After performing the search, these were the best parameters found from the search space ('max_depth': 10, 'n_estimators': 50).

The ensemble nature of RF allows to make it robust to overfitting and able to handle high-dimensional data with multiple features while being lightweight. The Deep Neural Network (DNN) model is more capable to capture complex, non-linear relationships in the data. The hyperparameters were selected (Number of layers: 3, Activation Function: ReLU, Optimizer: Adam, Batch Size: 32, Dropout Rate: 0.3, Learning Rate = 0.001), Dropout Layers, Early Stopping, and Regularization were included to ensure that overfitting does not occur. The models were deployed on the edge server and were fed features that were transmitted from the edge devices.

E. Integration of LLM in Cloud

In the proposed system, a Large Language Model (LLM) is hosted on a cloud service and is utilized for the tasks

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"""" You are an AI assistant specializing in cybersecurity and network
configuration. Your task is to analyze network anomalies detected on edge
devices and modify Apache server configuration files to mitigate potential
cyber-attacks.
You will be provided with the following information:
1. Network packet features of the flagged anomaly
2. The class or type of the detected anomaly
3. Current Apache server configuration files

Your objective is to:
1. Analyze the provided information
2. Determine the appropriate modifications needed in the configuration
files to address the specific anomaly and prevent similar occurrences
3. Generate and return the modified configuration files

Input format:
Packet Features: [Relevant features of the flagged network packet]
Anomaly Type: [Classification or type of the detected anomaly]
Current Configuration: [Relevant Apache server configuration files]

Output format:
Modified Configuration: [Updated Apache server configuration files]
Explanation: [Brief justification for the changes made]
Your goal is to enhance the security of the edge devices while maintaining
their functionality.""""
```

Fig. 3. The prompt provided to Claude 3.5 Sonnet.

of mitigation of any anomaly. The LLM plays an important role because, with its contextual understanding ability, it can interpret the features of the network packet and the attack and, accordingly, modify the configurations of the edge devices. Its code generation capability is very crucial for this task and in automating resolution strategies during an active Distributed Denial of Service (DDoS) attack or other network anomalies. The LLM's roles include automation of configuration updates and providing appropriate response to anomalies. Hosting the LLM on the cloud provides many benefits that include Scalability, Real-time Adaptability, Automation Efficiency, and No resource constraints at local level among many others. The LLM response that includes the updated configuration was used to update the configurations of files on the edge devices utilizing automation tools such as Jenkins. This ensures that the whole system is automated and has minimal to no human interaction.

IV. Results & Discussion

For evaluation of the proposed system, we conducted a series of tests. A subset of the data set was considered based on available system resources; Table 1 shows the configurations of the systems used during the tests. The setup involved simulating DDoS attacks in a controlled environment on a test Apache2 server running on a Linux Virtual Machine. The attacks were performed using tools such as hping3 and Low Orbit Ion Cannon (LOIC); this is to simulate real-world DDoS attacks. Both the random forest classifier and the deep neural network were trained on the dataset and tested. The Random Forest Classifier gave a train accuracy of 99.99% and a test accuracy of 99.98%. It performed well as the data set consisted of three classes and had simple features. The DNN on similar lines performed well and gave an accuracy of around 99%. Fig. 5 shows the training and validation accuracy of the DNN model. The graphs show similarity to models that might be over-fitting, but all the necessary precautions were taken to avoid over-fitting. Although similar models were used by other papers, their models were trained on the entire dataset. Given the reduced complexity of the problem, DNN may be too complex and excessive for this specific task, leading to the observed behavior in the graphs. Simpler architectures were tested, and we simplified down from complex models to the three-layered network presented in the paper, but the graphs still showed this pattern.

These metrics show the model's ability to detect DDoS attacks while minimizing false positives. In addition to deep learning anomaly detection, the system used Claude 3.5 Sonnet LLM through API calls to assist in automating the resolution

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TABLE I
SYSTEM CONFIGURATIONS

Component	Edge Device	Server
CPU	11th gen Intel i7-11800H	AMD Ryzen Threadripper Pro
GPU	Nvidia RTX 3050 Mobile	Nvidia Quadro RTX 4000
Memory	16 GB	64 GB
No. of Cores	8	24

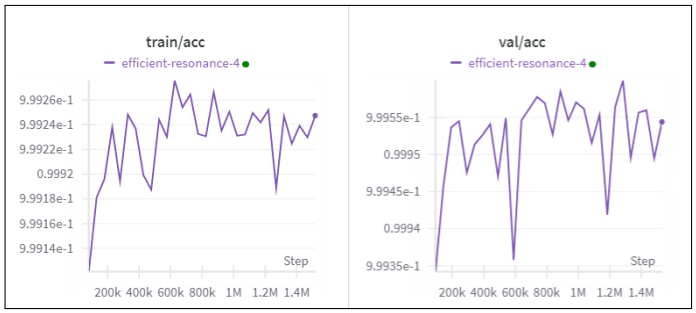

Fig. 4. Training and Validation Accuracy.

of the anomalies and updating of the configuration files. Fig. 3 shows the prompt used to send files and network data. During test attacks, the models were able to successfully detect the DDoS attacks and prompt the LLM. The LLM in response gave the appropriate responses on multiple occasions. For example, in response to an HTTP attack, the LLM responded with an updated configuration file that included code to block the specific IP address from which the packets originated. And with the use of automation tools such as Jenkins, the entire system was automated.

The current evaluation was performed with a single edge server setup. Although this configuration was sufficient for validating the system's functionality and performance, we acknowledge that production environments would require a distributed architecture. In real-world deployments, the system can be scaled horizontally by implementing multiple edge servers with load balancing mechanisms. This distributed approach would enable handling of a significantly larger number of edge devices. Future work could include testing and benchmarking this distributed configuration to evaluate its performance.

We acknowledge that the edge device specifications used in our experimental setup are more powerful than typical resource-constrained edge devices in production environments. This hardware was used primarily for development and proof-of-concept testing. However, our client-edge co-inference approach is specifically designed to be compatible with less powerful edge devices. By distributing the computational workload where the edge devices only perform initial network traffic analysis and feature extraction, while the edge server handles the resource-intensive machine learning inference, our system reduces the burden on the edge devices. This approach enables our solution to function effectively on standard edge computing devices with minimal specifications. Future work will include validation testing on such resource-constrained hardware to confirm performance in realistic edge deployment cases.

V. CONCLUSION AND FUTURE WORK

The study demonstrates the effectiveness of the Edge AI-based system that integrates LLM. The system was able to perform automatically with minimal human intervention. It performs better than traditional methods by providing faster detection and response times due to real-time anomaly detection at the edge. This enhances scalability. The architecture also shows potential beyond just DDoS detection. A similar architecture can be adapted for various different types of intrusion detections and also for IoT environments. The integration of LLMs allows for automated responses and adaptability to different types of attacks, future applications hold great potential with the increase in LLM's capabilities and access to more powerful resources within the commodity space. Applications may include predictive maintenance in smart manufacturing, assistance in dynamic policy adjustments, real-time troubleshooting, and a wide variety of other tasks beyond security. The current system could further be improved with the use of fine-tuning techniques or a retrieval Augmented Generation (RAG) based system to make the system more suitable for the task at hand.

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